

Clarification on Specimen Design and Coursework Requirements

1. Designing Your Specimen:

- You do not need to create a specimen that exactly looks like the final product of your chosen application. Instead, you are required to modify the given ASTM D638 specimen in specific ways that suit the practical requirements of your chosen application.
- For example:
 - In an Aerospace application, you might extend the grip areas of the specimen to enhance load distribution in high-stress environments.
 - In a Medical application, you could adjust the thickness of the specimen to replicate orthopaedic supports for improved structural integrity.
- The key is to focus on modifications in the design that align with the needs of your application, and to justify these design choices in your report using engineering principles.

2. Materials for 3D Printing:

- We will be using the Tiertime UP Box+ 3D Printer that are available in the QMES practical laboratory, which is limited to printing only polymer materials (specifically ABS and PLA).
- Even if your application traditionally requires metallic materials, you will only be using ABS and PLA for this task.

3. Modified Specimen:

- You are only required to create and modify one specimen, which will be 3D printed separately using i) ABS and ii) PLA
- The same specimen should be printed using both materials to allow you to compare how the properties of ABS and PLA affect its behavior and performance.

4. Testing Your Specimen:

- You will need to use your engineering knowledge and judgment to determine appropriate testing methods for your modified specimen.
- Focus on testing deformation and failure under conditions relevant to your modified specimen and application, but the specifics of the test design are up to you, as long as they are based on sound engineering principles.